Earth's Revolution: Seasons, the Ecliptic, Equinoxes and Solstices

What you will be able to do

1. Explain the real cause of the seasons — and bust the "distance" myth.
2. Define the ecliptic, the zodiac, the equinoxes and the solstices.
3. Calculate the Sun's noon altitude for any Indian city on key days.
4. Predict your city's **Zero Shadow Day**.
5. Explain why **Makar Sankranti** falls on 14 January and not on the winter solstice (21 December), and connect solar festivals across India to astronomy.

1. Earth's journey around the Sun

Quantity	Value
Average distance from the Sun	149.6 million km = **1 astronomical unit (AU)**
Orbital speed	$\approx 29.8\,\text{km/s} \approx 1{,}07{,}000\,\text{km/h}$
One orbit relative to the stars (*sidereal year*)	365.256,4 days
March equinox to March equinox (*tropical year* — the year of the seasons)	365.242,2 **days**
Closest to the Sun (**perihelion**)	~3 January — 147.1 million km
Farthest from the Sun (**aphelion**)	~4 July — 152.1 million km

Myth-buster

Earth is *closest* to the Sun in early January — in the middle of the northern winter. So distance is **not** the cause of the seasons. (The 3% change in distance makes only a small difference.)

2. The ecliptic: the Sun's path among the stars

As Earth orbits, the Sun appears to move slowly **eastward** against the background stars,

$$\frac{360^\circ}{365\text{ days}} \approx 1^\circ \text{ per day.}$$

We cannot see the stars beside the Sun in daylight, but we can notice that the star patterns seen just after sunset change month by month.

- The yearly path of the Sun across the sky is called the **ecliptic**.
- The Moon and the planets always stay close to the ecliptic, because the solar system is quite flat.
- The band of star patterns along the ecliptic is the **zodiac**. In Indian astronomy it is divided into 12 **rashis**; the Moon's path is also divided into 27 **nakshatras**.

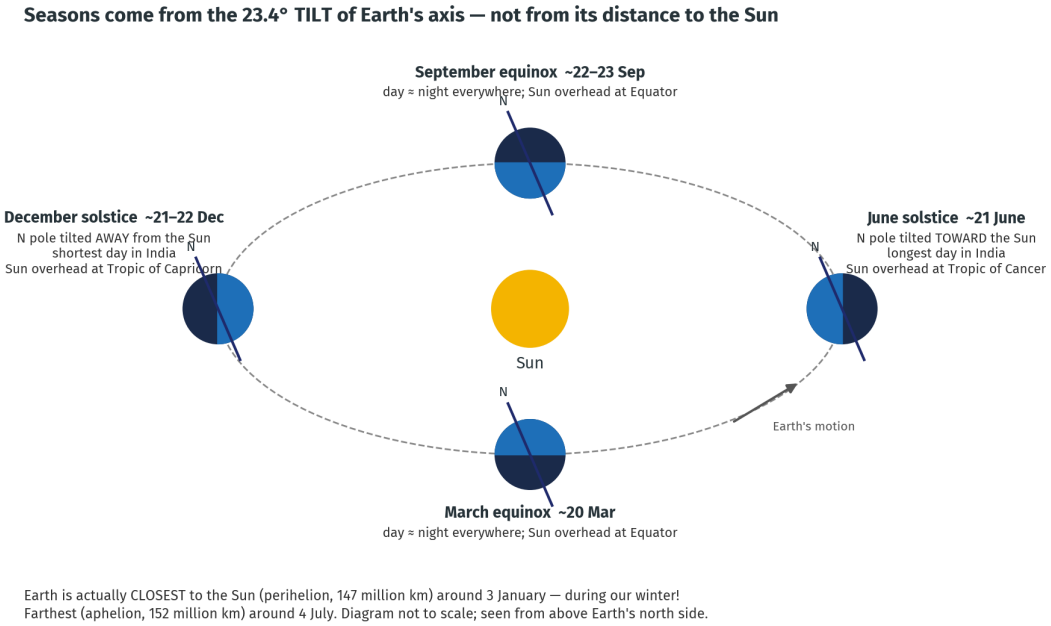
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P.T.O.

The tilt that makes everything happen

Earth's axis is tilted by $\varepsilon = 23.44°$ from the perpendicular to its orbit (the **obliquity of the ecliptic**). As a result, the ecliptic is tilted 23.44° to the **celestial equator** — the projection of Earth's equator onto the sky.

3. The real reason for the seasons



Earth's axis always points the same way in space (towards Polaris). In June the northern hemisphere leans towards the Sun; in December it leans away.

The tilt has two effects:

1. **The angle of sunlight.** When the Sun is high, its rays fall almost straight down and the energy is concentrated on a small area — it is hot. When the Sun is low, the same light spreads over a larger area — it is cooler. (Try it with a torch shining on a table.)
2. **The length of the day.** In summer the Sun is above the horizon for longer: more hours of heating, fewer of cooling.

4. The four turning points of the year

Event	Approx. date	Sun's dec.	What happens
March equinox	20–21 Mar	0°	Sun crosses the celestial equator going north; day ≈ night everywhere; Sun rises **due east**.
June solstice	20–21 Jun	+23.44°	Sun farthest north; longest day in India; overhead at the **Tropic of Cancer**.
September equinox	22–23 Sep	0°	Sun crosses the equator going south; day ≈ night.
December solstice	21–22 Dec	−23.44°	Sun farthest south; shortest day in India; overhead at the **Tropic of Capricorn**.

Word origins. *Equinox* = *aequus* (equal) + *nox* (night). *Solstice* = *sol* (Sun) + *sistere* (to stand still) — near the solstices the Sun's rising point on the horizon seems to stop and turn back.

Indian connection — the Tropic of Cancer runs through India

It crosses **8 states**: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram. On the June solstice the noon Sun is exactly overhead along this line.

The Sun's declination through the year

Declination (δ) is the Sun's angle north ($+$) or south ($-$) of the celestial equator. Values for 2027 (they repeat closely every year):

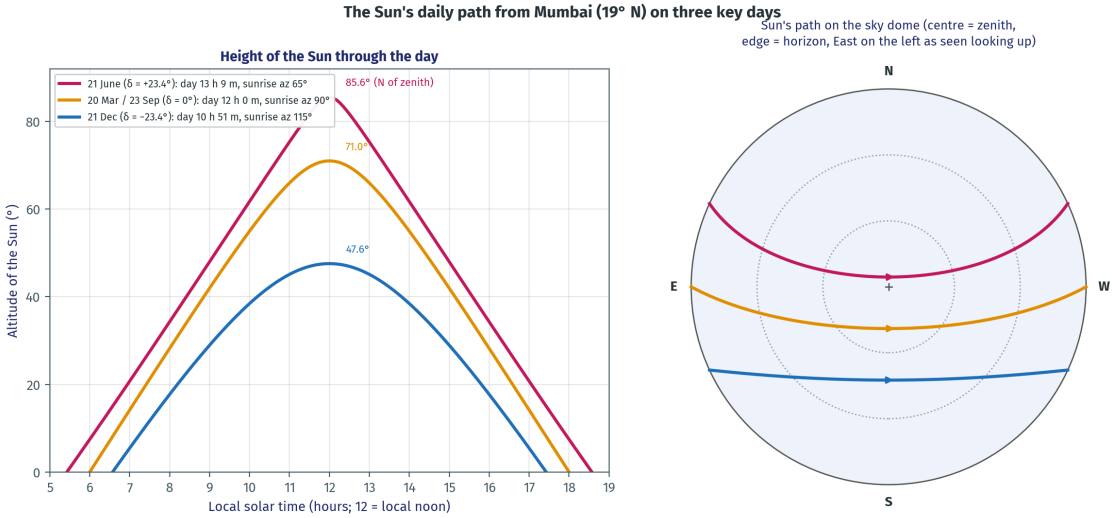
Date	δ	Date	δ	Date	δ
1 Jan	$-23.0°$	1 May	$+14.9°$	1 Sep	$+8.5°$
1 Feb	$-17.3°$	1 Jun	$+22.0°$	23 Sep	$0.0°$
1 Mar	$-7.8°$	21 Jun	$+23.4°$	1 Oct	$-3.0°$
21 Mar	$0.0°$	1 Jul	$+23.1°$	1 Nov	$-14.2°$
1 Apr	$+4.3°$	1 Aug	$+18.2°$	1 Dec	$-21.7°$
14 Apr	$+9.2°$			22 Dec	$-23.4°$

5. How high is the Sun at noon?

At local noon the Sun crosses the *meridian* — the north–south line passing directly overhead. Its altitude then is

$$\boxed{h_{\text{noon}} = 90° - \varphi + \delta}$$

where φ is your latitude and δ the Sun's declination. If the result is greater than $90°$, the Sun is **north** of the point overhead and its altitude is $180° - h_{\text{noon}}$.



The Sun's daily path seen from Mumbai on three key days. In June the noon Sun is actually slightly north of overhead.

Worked examples

1. Delhi ($\varphi = 28.6°$) on the June solstice: $h = 90° - 28.6° + 23.4° = \mathbf{84.8}°$ — almost overhead, but still in the south.
2. Delhi on the December solstice: $h = 90° - 28.6° - 23.4° = \mathbf{38.0}°$ — a low winter Sun.

Worked examples *(continued)*

3. Mumbai ($\varphi = 19.1°$) on the June solstice: $h = 90° - 19.1° + 23.4° = 94.3° > 90°$, so the Sun is north of overhead at $180° - 94.3° = \mathbf{85.7}°$. Noon shadows point **south**!

Where does the Sun rise?

Only at the equinoxes does the Sun rise exactly due east. In June it rises towards the north-east, in December towards the south-east. From Mumbai the sunrise point swings between about 65° and 115° of azimuth — a 50° swing along the horizon in a year. Many ancient monuments were aligned to these points.

6. Zero Shadow Day

Everywhere between the two Tropics, the noon Sun is **exactly overhead twice a year**, on the dates when

$$\delta_{\text{Sun}} = \varphi_{\text{your city}}.$$

At that moment vertical objects cast **no shadow**. Most of India lies south of the Tropic of Cancer, so most Indians can see this.



Students in Tamil Nadu observing Zero Shadow Day: the upright objects on the ground cast almost no shadow at local noon.

Photo: TNSC Shanmu VPM, CC BY-SA 4.0, via Wikimedia Commons

City	Latitude	Zero Shadow Days (approx., at local noon)
Kanyakumari	8.1° N	11 April & 2 September
Thiruvananthapuram	8.5° N	12 April & 1 September
Bengaluru / Chennai	13.0° N	25 April & 19 August
Hyderabad	17.4° N	10 May & 4 August
Pune	18.5° N	14 May & 31 July
Mumbai	19.1° N	16 May & 28 July
Bhubaneswar	20.3° N	22 May & 22 July

(continued)

City	Latitude	Zero Shadow Days (approx., at local noon)
Nagpur	21.2° N	27 May & 18 July
Kolkata	22.6° N	6 June & 8 July
Ahmedabad	23.0° N	11 June & 3 July
Delhi	28.6° N	**Never** — north of the Tropic of Cancer

Computed for 2027; dates shift by ± 1 day from year to year. "Noon" means local solar noon, which is not exactly 12:00 IST.

Activity

On your city's Zero Shadow Day, stand a straight stick or bottle upright in the sunshine around local noon and watch its shadow shrink and vanish beneath it. The Astronomical Society of India runs public Zero Shadow Day campaigns every year.

7. Makar Sankranti: the festival that tells an astronomical story

What is a *Sankranti*?

A **Sankranti** is the moment the Sun moves from one **rashi** (zodiac sign) into the next. There are 12 Sankrantis a year — about one a month. Two are celebrated almost everywhere in India:

Sankranti	Date	Festivals across India
Makar Sankranti — Sun enters Makara	**14–15 Jan**	Makar Sankranti (Maharashtra, Karnataka); **Pongal** (Tamil Nadu); **Uttarayan** kite festival (Gujarat); **Lohri** on the eve (Punjab); **Magh Bihu** (Assam); Khichdi (UP, Bihar); Poush Sankranti (West Bengal); Maghi (Punjab)
Mesha Sankranti — Sun enters Mesha	**13–14 Apr**	**Vaisakhi** (Punjab); **Vishu** (Kerala); **Puthandu** — Tamil New Year; **Bohag Bihu** (Assam); **Pohela Boishakh** — Bengali New Year; Pana Sankranti (Odisha); Jur Sital (Mithila)

These are **solar** festivals, so their Gregorian dates hardly change from year to year — unlike festivals that follow the Moon.

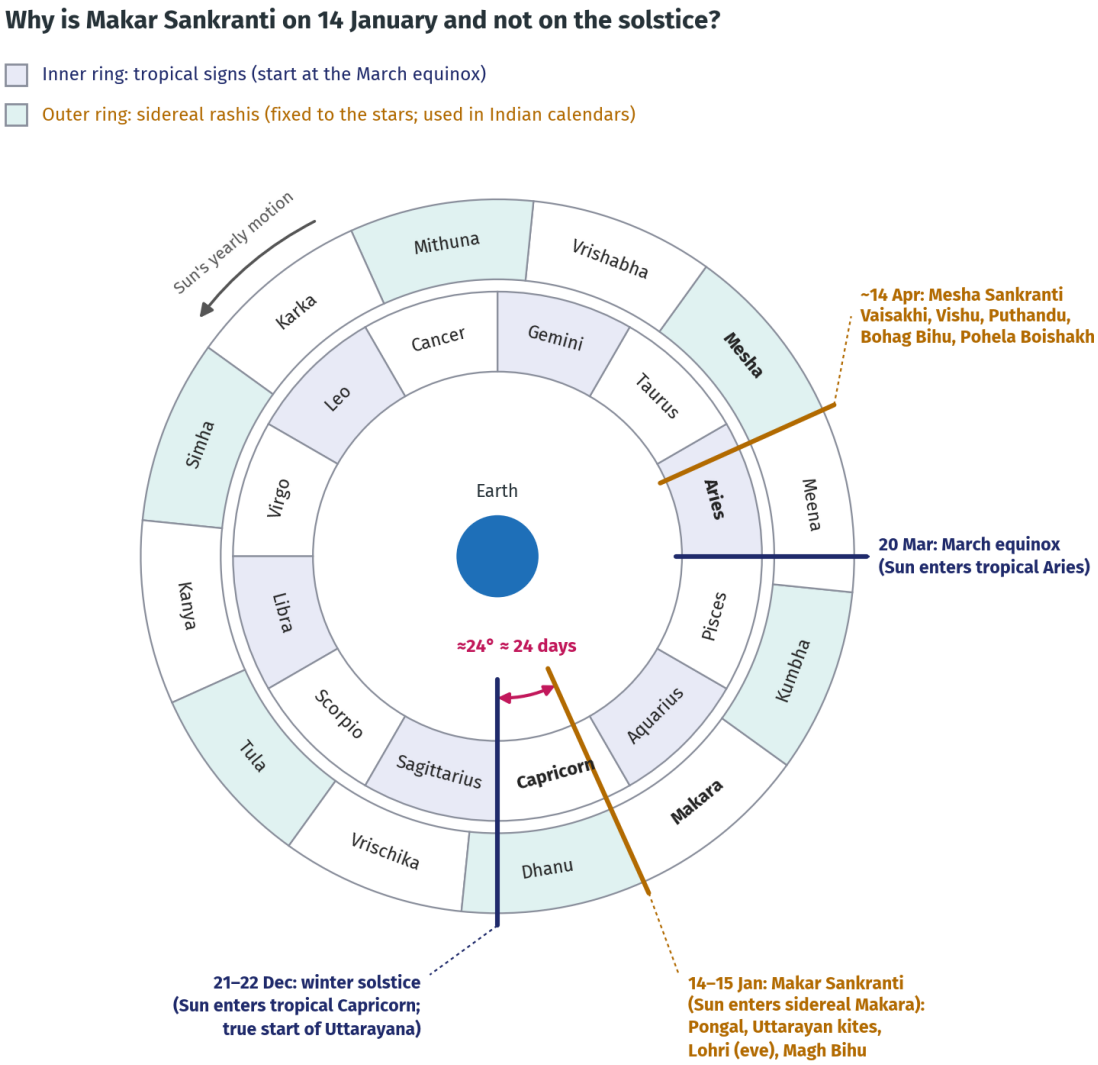
Uttarayana: the Sun's "northward journey"

After the winter solstice, the Sun's rising point starts moving north again and the days grow longer. This six-month period is called **Uttarayana**; the other half of the year is **Dakshinayana**. Astronomically, Uttarayana begins at the **winter solstice, 21–22 December**. So why is it celebrated on 14 January?

The answer: precession and the two zodiacs

- Earth's axis slowly wobbles like a spinning top, tracing one circle in about 25,800 years — the **precession of the equinoxes**, discovered by the Greek astronomer Hipparchus (~130 BCE).
- So the equinox points slide **westward** along the zodiac by

$$\frac{360^\circ}{25{,}800\ \text{yr}} \approx 50.3'' \text{ per year} \approx 1^\circ \text{ every 72 years.}$$



Why Makar Sankranti and the winter solstice are about 24 days apart.

- The **tropical zodiac** (used in the West) starts at the March equinox, so it slides with precession.
- The **sidereal zodiac** (*nirayana*, used in Indian calendars) is fixed to the **stars**.
- About 1,700 years ago the two zodiacs lined up. Since then they have drifted about **24**° apart — the **ayanamsha**.

So the Sun reaches the start of *tropical* Capricorn at the solstice (21 December) but the start of *sidereal* Makara about 24 days later (14 January). Makar Sankranti is the day the Sun enters the star-based Makara.

A prediction you can make

Since 1° ≈ 1 day of the Sun's motion and 1° of precession takes ~72 years, Makar Sankranti drifts about **1 day later every ~70 years**. In the early 1900s it often fell on 13 January; today it is 14–15 January; in about 13,000 years it will fall in **July**!

Precession also changes the Pole Star

The axis now points near Polaris, which will be closest to the pole around the year 2100. About 4,800 years ago the pole star was **Thuban** (in Draco); in about 12,000 years it will be **Vega** (*Abhijit*).

Try this week

1. **Sunset tracker.** Every week, photograph the sunset from the same spot and mark where the Sun sets against buildings or trees. Over 2–3 months you will see it move along the horizon.
2. **Shadow stick.** Fix a vertical stick in open ground. Mark the tip of its shadow every 15 minutes around noon. The shortest shadow points along the north–south line, and its time is your **local noon**.

Key points to remember

- Seasons are caused by the 23.44° **tilt**, not by distance; Earth is **closest to the Sun in January**.
- **Ecliptic** = the Sun's yearly path; **zodiac** = the star patterns along it.
- Equinoxes ~20 March and ~22–23 September; solstices ~21 June and ~21–22 December.
- $h_{\text{noon}} = 90^\circ - \varphi + \delta$.
- Places south of 23.44° N have **two Zero Shadow Days** a year.
- **Makar Sankranti** (14 Jan) = Sun's entry into **sidereal Makara**; the winter solstice (21 Dec) is the true start of Uttarayana; the ~24-day gap is due to **precession**.

Check yourself

1. Why is it hotter in May than in December in India, even though Earth is farther from the Sun in May?
2. What is the noon altitude of the Sun in Kolkata (22.6° N) on the March equinox?
3. Name three festivals in three different states that fall at Mesha Sankranti.
4. Will Chandigarh (30.7° N) ever have a Zero Shadow Day? Why?

Answers: 1. The Sun is higher and days are longer. 2. $90^\circ - 22.6^\circ + 0^\circ = 67.4^\circ$. 3. E.g. Vaisakhi (Punjab), Vishu (Kerala), Bohag Bihu (Assam). 4. No — it is north of the Tropic of Cancer.

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §4.2 "The Seasons" — https://openstax.org/books/astronomy-2e/pages/4-2-the-seasons
- Astronomical Society of India, Public Outreach & Education Committee — Zero Shadow Day resources: https://astron-soc.in/outreach/
- NASA Space Place, "What causes the seasons?" — https://spaceplace.nasa.gov/seasons/